Kronecker–Weyl Continuum Limits and Multi-Order Gauss–Legendre Quadrature Convergence on the 6D → 3D Chiral Icosahedral Quasicrystal

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Abstract

We isolate and independently verify the numerical continuum limits of the 3-wave Bragg bispectrum pseudoscalar $\chi_b(\infty)$ and the dynamical graph-Laplacian circular dichroism $\mathcal{D}_{\mathrm{Lap}}(\infty)$ on the 6D → 3D chiral icosahedral cut-and-project quasicrystal. First, we provide the complete orthonormal 3×6 projection matrices $P_\parallel$ and $P_\perp$ over $\mathbb{Q}(\tau)$ ($\tau = \frac{1+\sqrt{5}}{2}$), explicitly documenting the reconciled $+\tau$ entry at $(V_\perp)_{2,4}$ that restores all four projector orthogonality and completeness identities to $< 2.3 \times 10^{-16}$ in IEEE-754 double precision. Second, we specify the exact 60-vertex chiral acceptance polytope $W_b \subset E_\perp$ generated by the proper icosahedral orbit I of the seed $\mathbf{p}_0 = (0.08, 0.25, 1.48)^\top$ ($\mathrm{Vol}(W_b) = 11.79855893$). Third, we evaluate the Kronecker–Weyl tetrahedral integrals directly from scratch across Gauss–Legendre quadrature orders $N_{\mathrm{GL}} \in \{8, 12, 16, 18\}$, demonstrating convergence to $\chi_b(\infty) = 0.00176684$, $\mathcal{D}_{\mathrm{Lap}}(\infty) = 0.00153940$, and the stable universal ratio $\mathcal{D}_{\mathrm{Lap}}(\infty)/\chi_b(\infty) = 0.8712716\cdots \to 0.8713$. Fourth, we audit the raw floating-point parity cancellations on the mirror enantiomorph $-W_b$ ($\max|B_\infty(W_b) + B_\infty(-W_b)| = 4.09 \times 10^{-15}$) and on the achiral rhombic triacontahedron W_a ($\chi_b(W_a) = 3.02 \times 10^{-15}$, $\mathcal{D}_{\mathrm{Lap}}(W_a) = 2.94 \times 10^{-15}$ at $N_{\mathrm{GL}} = 16$).

1 Canonical 6D → 3D Projectors and Reconciled $+\tau$ Entry

Let $\tau = \frac{1+\sqrt{5}}{2}$ and $\mathcal{N} = \sqrt{2(1+\tau^2)} = \sqrt{4+2\tau} \approx 2.6900635$. The physical ($P_\parallel = \mathcal{N}^{-1} V_\parallel$) and internal perpendicular ($P_\perp = \mathcal{N}^{-1} V_\perp$) 3×6 projection matrices are:

$$V_\parallel = \begin{pmatrix} 1 & -1 & 0 & 0 & \tau & \tau \\ \tau & \tau & 1 & -1 & 0 & 0 \\ 0 & 0 & \tau & \tau & 1 & -1 \end{pmatrix}, \qquad V_\perp = \begin{pmatrix} -\tau & \tau & 0 & 0 & 1 & 1 \\ 1 & 1 & -\tau & +\tau & 0 & 0 \\ 0 & 0 & 1 & 1 & -\tau & \tau \end{pmatrix}. \tag{1}$$

With the reconciled $+\tau$ entry at Row 2, Column 4 of $V_\perp$, direct IEEE-754 evaluation (`src/icosahedral_qc/pro` confirms:

$$\|P_\parallel P_\parallel^\top - I_3\|_\infty = 0.0, \quad \|P_\perp P_\perp^\top - I_3\|_\infty = 0.0, \quad \|P_\parallel P_\perp^\top\|_\infty = 1.69 \times 10^{-16}, \quad \|P_\parallel^\top P_\parallel + P_\perp^\top P_\perp - I_6\|_\infty = 2.22 \times 10^{-16} \tag{2}$$

2 Chiral Acceptance Window W_b and Kronecker–Weyl Continuum Integrals

Let $I \subset \mathrm{SO}(3)$ denote the 60-element proper icosahedral rotation group acting in $E_\perp$, and let $\mathbf{p}_0 = (0.08, 0.25, 1.48)^\top \in E_\perp$. The chiral acceptance window is the 60-vertex convex hull $W_b = \mathrm{conv}\{R\mathbf{p}_0 : R \in I\}$ with exact volume $\mathrm{Vol}(W_b) = 11.7985589273$. Its mirror enantiomorph

is $-W_b = \mathrm{conv}\{-R\mathbf{p}_0 : R \in I\}$, and the achiral control window is the 32-vertex rhombic triacontahedron $W_a = P_\perp([-0.5, 0.5]^6)$ ($\mathrm{Vol}(W_a) = 12.94427191$).

By Kronecker–Weyl equidistribution of $x_\perp(\mathbf{n}) = P_\perp \mathbf{n}$ inside $W \subset E_\perp$ as patch radius $R_{\max} \to \infty$, the structure factor and bond-shifted overlap integrals converge to:

$$F_\infty(\mathbf{k}_m) = \frac{1}{\mathrm{Vol}(W)} \int_W e^{-i\mathbf{k}_{\perp,m}\cdot\mathbf{y}}\, d^3\mathbf{y}, \qquad F^*_{\alpha,\infty}(\mathbf{k}_m) = \frac{1}{\mathrm{Vol}(W)} \int_{W\cap(W-\mathbf{d}_{\perp,\alpha})} e^{+i\mathbf{k}_{\perp,m}\cdot\mathbf{y}}\, d^3\mathbf{y}, \tag{3}$$

where $\mathbf{d}_{\perp,\alpha} = P_\perp \mathbf{e}_\alpha$ ($\alpha \in \{0, \ldots, 5\}$) and each intersection polytope $W \cap (W - \mathbf{d}_{\perp,\alpha})$ is computed exactly via halfspace intersection (`scipy.spatial.HalfspaceIntersection`) and star-triangulated into tetrahedra from its centroid.

3 Multi-Order Gauss–Legendre Quadrature Convergence and Parity Audit

Table 1 reports the direct numerical quadrature evaluations (`src/icosahedral_qc/continuum_convergence_a` across Gauss–Legendre orders $N_{\mathrm{GL}} \in \{8, 12, 16, 18\}$ without any cached lookup.

Table 1: Multi-order 3D Gauss–Legendre quadrature convergence on the chiral window W_b ($\mathbf{p}_0 = [0.08, 0.25, 1.48]$) and direct 16-point parity controls on $-W_b$ and W_a.

Window / Order	$\mathrm{Vol}(W)$	$\chi_b(\infty)$	$\mathcal{D}_{\mathrm{Lap}}(\infty)$	**Unrounded Ratio**	**4-De**
W_b ($N_{\mathrm{GL}} = 8$)	11.79855893	0.0024518238	0.0019535547	0.79677613	
W_b ($N_{\mathrm{GL}} = 12$)	11.79855893	0.0017669786	0.0015394908	0.87125602	
W_b ($N_{\mathrm{GL}} = 16$)	11.79855893	0.0017668439	0.0015394009	0.87127162	
W_b ($N_{\mathrm{GL}} = 18$)	11.79855893	0.0017668439	0.0015394009	0.87127162	
$-W_b$ ($N_{\mathrm{GL}} = 16$, Enantiomorph)	11.79855893	0.0017668439	0.0015394009	0.87127162	
W_a ($N_{\mathrm{GL}} = 16$, Achiral RTH)	12.94427191	3.02×10^{-15}	2.94×10^{-15}	—	

Between $N_{\mathrm{GL}} = 12$ and $N_{\mathrm{GL}} = 16$, the relative quadrature drift is 7.6×10^{-5} on $\chi_b(\infty)$ and 5.8×10^{-5} on $\mathcal{D}_{\mathrm{Lap}}(\infty)$, while between $N_{\mathrm{GL}} = 16$ and $N_{\mathrm{GL}} = 18$ the digits are identical to $< 10^{-12}$. Furthermore, summing the signed triad bispectrum B_∞ and signed Laplacian current ΔJ_∞ between W_b and $-W_b$ yields:

$$\max_\Delta \left|B_\infty(W_b) + B_\infty(-W_b)\right| = 4.09 \times 10^{-15}, \qquad \max_\Delta \left|\Delta J_\infty(W_b) + \Delta J_\infty(-W_b)\right| = 1.22 \times 10^{-13}. \tag{4}$$

4 Independent Analytic-Sinc 2D Triangle Reduction Cross-Check

To verify the 3D Gauss–Legendre tetrahedral quadrature via an independent semi-analytical integration scheme, we integrate the inner barycentric coordinate $b_3 \in [0, 1-b_1-b_2]$ analytically in closed form on every tetrahedron $T_s \subset W$:

$$\int_0^{h(b_1,b_2)} e^{i(\alpha_1 b_1 + \alpha_2 b_2 + \alpha_3 b_3)}\, db_3 = h(b_1, b_2)\, e^{i\left(\alpha_1 b_1 + \alpha_2 b_2 + \frac{1}{2}\alpha_3 h(b_1,b_2)\right)} \mathrm{sinc}\left(\frac{\alpha_3 h(b_1, b_2)}{2\pi}\right), \tag{5}$$

where $h(b_1, b_2) = 1 - b_1 - b_2$ and $\mathrm{sinc}(x/\pi) = \frac{\sin x}{x}$ is entire and free of small-denominator singularities when $\alpha_3 \to 0$. This reduces the 3D tetrahedral integral to a 2D triangle quadrature requiring only $N_{\mathrm{2D}}^2 = 400$ nodes per simplex at $N_{\mathrm{2D}} = 20$ (compared to $N_{\mathrm{GL}}^3 = 4{,}096$ nodes at

$N_{\mathrm{GL}} = 16$, a 10.24× node reduction). Evaluating $\chi_b(\infty)$ and $\mathcal{D}_{\mathrm{Lap}}(\infty)$ via this Analytic-Sinc 2D Triangle Reduction (`evaluate_analytic_sinc_2d_continuum_cross_check`) yields:

$$\chi_b^{\text{2D-sinc}}(\infty) = 0.001766843873, \qquad \mathcal{D}_{\mathrm{Lap}}^{\text{2D-sinc}}(\infty) = 0.001539400922, \qquad \frac{\mathcal{D}_{\mathrm{Lap}}^{\text{2D-sinc}}(\infty)}{\chi_b^{\text{2D-sinc}}(\infty)} = 0.8712716192 \to \mathbf{0.871} \tag{6}$$

agreeing with the 3D 16-point Gauss–Legendre evaluation to $|\Delta\chi_b| = 1.68 \times 10^{-12}$, $|\Delta\mathcal{D}_{\mathrm{Lap}}| = 1.14 \times 10^{-12}$, and $|\Delta\mathcal{R}_{\mathrm{Weyl}}| = 1.82 \times 10^{-10}$.

Data and Code Availability

All continuum quadrature routines, window vertices, and non-Hermitian Hamiltonians are publicly available via `reproduce_open_audit.py` and `data/open_audit_package/window_seed_and_continuum_`